

Geo-spatial modelling of vehicle crime in Northern Ireland using computer vision to identify environmental factors.

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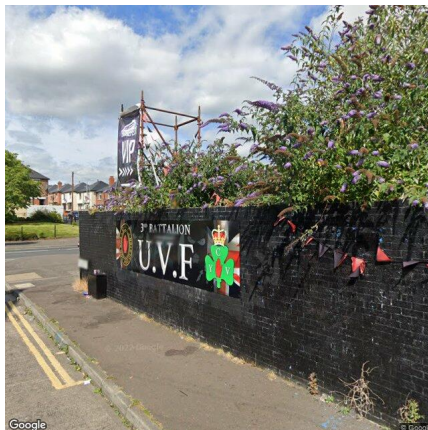
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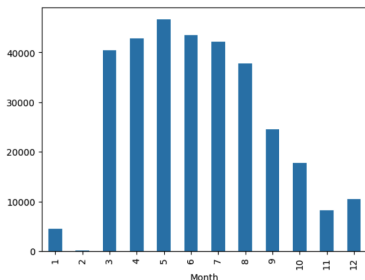
Background

- “The Troubles” (1960’s - 1998), Northern Ireland.
- Identifying paramilitary flags/murals.

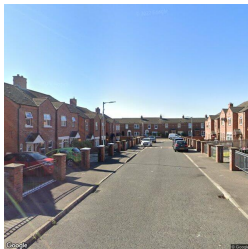


GSV Images

- 1.9 million GSV images for Northern Ireland, in particular 332,490 images for Belfast City.
- 55,415 unique locations within this area.
- Between July 2008 and July 2023.



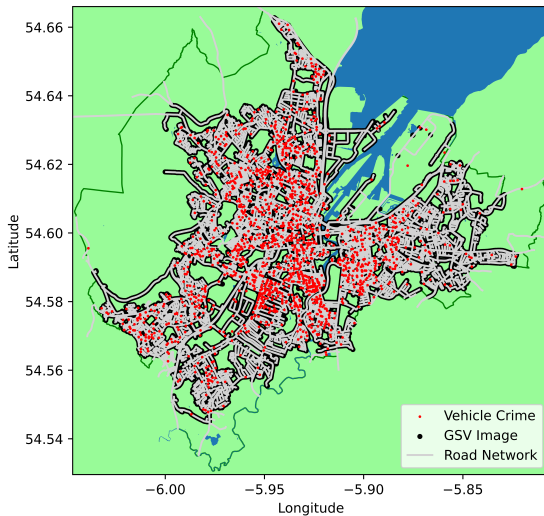
GSV Example



Combining Data Sources

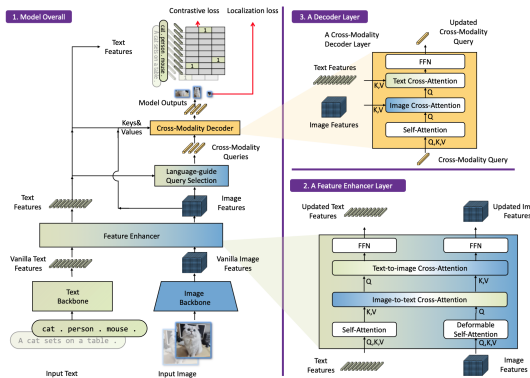
- Vehicle crime data from <https://data.police.uk> for Northern Ireland.
- 2,700 vehicle crimes took place in Belfast City between September 2020 and July 2023.
- Road networks and maps from OpenStreetMap.

Belfast City



GroundingDINO

- GroundingDINO [1] is an open-set object detector that combines DINO [3], a transformer-based detector, with grounded pre-training.



What to look for

Enablers and Deterrents to Crime

- Signs of decay and dilapidation, e.g. broken windows, abandoned buildings, vacant sites, overgrown vegetation, graffiti and litter.
- Presence of police, amenities, self-policing communities (through paramilitary flags and murals) and surveillance devices.

Cars	PP	PN
P	1383	179
N	170	73

F1 Score	0.8879
Precision	0.8905
Recall	0.8854
RMSE	1.2145
MAE	0.5817

Object Detection Example



Actuarial Applications

Replacement for Postal Codes

- Assumes historical data is a suitable indicator for future claims.
- Changes to the environmental factors may change claim rate but take time to filter through.
- This also addresses the reasons why certain areas are riskier than others.
- These changes can be picked up by GSV images.

Environmental changes



- With GSV images we can see the gradual regeneration of derelict sites.



Regression Modelling

- We model the number of vehicle crimes in a location using a Poisson regression[2]. The model takes the form,

$$\log(\mathbb{E}[Y|x]) = X\beta$$

where Y represents the number of vehicle crimes, X is a matrix of regressors and β is a vector of coefficients.

- The number of vehicle crimes in a location increases with the presence of litter, shops, graffiti and cars, but decreases with the presence of street lights.
- Model Coefficients:

Car	Graffiti	Light	Litter	Shops
1.0698	1.2623	0.6745	1.4479	1.2860

Continuation of Work

- Gemini - Google
- Confounding Variables, e.g. socio-economic status
- Premium implications via a simple notional product
- Spatial Clustering
- Time Series element

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